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Electricity user behavior analysis and marketing strategy based on internet of things and big data

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Abstract

This paper examines power user behavior and the design of marketing strategies, using a case study of Smart Community A. We explore how advanced analytical models are used to enhance energy efficiency and user services. First, we apply spectral clustering to refine user segmentation and identify distinct electricity consumption patterns among different groups. Then, the Hidden Markov Model (HMM) analyzes user behavior, uncovering shifts in consumption habits and enabling personalized service offerings. Next, the ARIMA model predicts electricity consumption trends, guiding grid scheduling and resource allocation. Based on these analyses, we develop targeted marketing strategies, such as dynamic pricing and energy-saving incentives, which boost user engagement and reduce energy usage. Through an IoT and big data-driven interactive marketing platform, we enhance user experience and foster a culture of energy conservation. Finally, a feedback mechanism ensures continuous improvement and maximizes the effectiveness of the marketing strategies.

Keywords: Internet of things, Big data, Power users, Behavioral analysis, Marketing strategy

Introduction

With the rapid development of information technology, the Internet of Things (IoT) and Big Data have become the key forces driving change in various industries. In the modern power system, the integration of these two technologies provides unprecedented opportunities for the intelligent transformation of the power industry. The purpose of this paper is to discuss in depth the analysis of power user behavior based on IoT and Big Data, and develop more accurate and effective marketing strategies on this basis, so as to not only improve the service efficiency of power enterprises and user satisfaction, but also have a far-reaching impact on the sustainable development of the entire energy system.

The Internet of Things (IoT) technology realizes a seamless connection between the physical world and the digital world through the deep integration of sensors, smart devices and the Internet. In the electric power system, the application of IoT has expanded from the initial remote monitoring and fault warning to all levels of

the smart grid. Smart meters, transmission line monitoring equipment, distributed energy management systems, etc., constitute a huge data collection network, real-time monitoring of grid operation status and user behavior. The real-time transmission and processing of these data enables the power system to respond more flexibly to various operating conditions, and improves the reliability and security of the power grid (Uhl et al. 2023).

Big data technology, on the other hand, gives electric power enterprises unprecedented data processing and analyzing capabilities. The daily operation of the power system generates a huge amount of data, which covers multiple dimensions such as users' power consumption, power consumption periods, load curves and so on. Through advanced data analysis techniques, such as machine learning and data mining, valuable information such as user behavior patterns and demand change trends can be distilled from this data. This insight means for power companies to be able to more accurately predict demand, optimize resource allocation, and design services that more closely match user needs (Zhai et al. 2020).

Recent advancements in the integration of IoT and big data technologies in the power industry have significantly enhanced operational efficiency and customer engagement. Recent studies highlight the use of IoT devices for real-time monitoring and predictive maintenance, reducing downtime and improving reliability. However, these works primarily focus on technical aspects and do not delve into the broader implications for customer-centric services (Wang et al. 2020). In terms of data analytics, recent research demonstrates the effectiveness of machine learning algorithms, specifically deep learning techniques, in forecasting electricity demand and optimizing grid operations. Despite these advances, there is a notable lack of comprehensive frameworks that integrate various data sources for holistic decision-making, indicating a significant gap in current research (Zhang et al. 2023). Critical analysis reveals that while many studies emphasize the benefits of IoT and big data, few address the challenges of data privacy and security. Furthermore, there is limited exploration of the scalability of these solutions across diverse geographical locations. These gaps in the literature underscore the need for more comprehensive research that addresses both the technical and social dimensions of IoT and big data integration in the power industry (Liu et al. 2019).

In the increasingly competitive power market environment, the user is no longer a passive recipient of services, but has become an active participant in market activities. Users' behavior directly affects the fluctuation of power demand, which in turn affects the operational efficiency of the power grid and the rationality of energy distribution.

Figure 1 illustrates a comprehensive research framework for power user behavior modeling, which consists of three interrelated key links: first, the data acquisition stage, followed by the core link of constructing the user behavior model, and finally, the link of realizing the feedback of user interaction. In this process, data acquisition constitutes the cornerstone of analysis and ensures a realistic foundation for model building; the development of the user behavior model serves as the pivot of the whole system, aiming at understanding and predicting users' electricity consumption habits in depth; and user interaction not only facilitates the practical application of the model, but also drives the model's continuous iteration and refinement through the collection of feedback information, forming a cyclical and self-improving process.

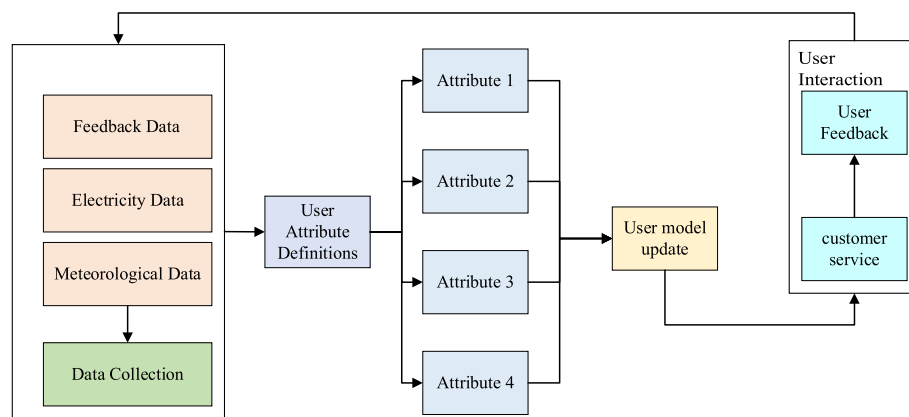


Fig. 1 Modeling of power user behavior

The core objective of this study is to reveal the behavioral patterns of power users and the motivation behind them with the power of IoT technology and big data analysis. By constructing a fine user portrait and identifying the differences in electricity consumption characteristics and demand of different user groups, it provides a scientific basis for electric power enterprises to design more accurate and personalized marketing strategies. This not only helps to improve user satisfaction and enhance user stickiness, but also further improves the overall efficiency of electric power services by optimizing resource allocation and reducing ineffective supply (Wang et al. 2020).

Marketing strategies based on user behavior analytics can help companies manage resources more effectively, reduce waste, and improve operational efficiency. For example, reducing outage time through predictive maintenance and balancing supply and demand through dynamic pricing strategies can reduce costs and increase revenue. Understanding user behavior enables power companies to respond quickly to changes in market demand, adjust service strategies, introduce products and services that meet user expectations, and enhance market competitiveness.

Literature review

The current status of the application of internet of things (IoT) technology in the electric power industry

In recent years, the Internet of Things (IoT) technology, like a trend that cannot be ignored, is deeply embedded in the bloodline of the power industry, and its influence is so great that it has fundamentally reshaped the operation and maintenance logic and management mode of the power system. Zhang et al. (2023) profoundly revealed the far-reaching impact of this technological innovation on power monitoring. Through the clever combination of intelligent sensors and advanced communication technology, the power system seems to be endowed with "senses", able to sense every pulse of the power grid in real time, whether it is voltage fluctuations, current anomalies, or small changes in equipment temperature, can be instantly captured and processed. This not only greatly improves the fault response speed, fault warning and accurate positioning, but also for the safe operation of the power grid to build a solid line of defense to ensure the continuity and stability of power supply.

The optimization of the data collection process by IoT technology has even given wings to the digital transformation of the power industry. Liu et al. (2019) points out that through the deployment of a wide range of IoT devices, huge amounts of field data can be transmitted to the cloud or data center in real time, forming an ocean of data. These data are not only large and diverse, providing valuable raw materials for subsequent data mining and analysis, but also providing the basis for scientific decision-making for electric power enterprises, and helping enterprises to realize the magnificent transformation from "experience-driven" to "data-driven".

In the field of equipment maintenance, predictive maintenance applications of IoT technologies have demonstrated their great potential in reducing O&M costs. This is elaborated in reference (Wu et al. 2023), where smart sensors monitor the status of equipment uninterruptedly and analyze and predict potential failures through algorithms, allowing maintenance to shift from reactive response to preventive action. Examples such as transformer oil temperature monitoring and mechanical vibration analysis have demonstrated the key role played by IoT technology in ensuring the long-term efficient and stable operation of power facilities.

Application of big data in user behavior analysis

The introduction of big data technology has opened a new window of insight into users for electric power enterprises. Wu et al. (2022) emphasized that by deeply mining big data on users' electricity consumption, enterprises can not only capture subtle trends in electricity demand, but also pinpoint the consumption habits of each user group, thus providing precise guidance for grid scheduling, avoiding the risk of overloading, and enhancing the flexibility and reliability of electricity supply. In addition, big data analysis makes personalized services possible. By identifying users' specific electricity consumption patterns, such as their preferences for peak and trough hours, power companies can design more reasonable time-sharing tariff strategies to effectively guide users to use electricity in a staggered manner, balance the load on the grid, and achieve efficient use of social resources. Luo et al. (2022) further added, by integrating social media data and user feedback, electric power companies can instantly grasp users' service satisfaction, quickly respond to market feedback, and continuously optimize the service experience, which is of inestimable value in enhancing customer loyalty and brand image.

Research related to behavioral analysis of electric power users

The basis of power user behavior consists of five elements: ① the subject is the cognitively capable user with socio-economic attributes; ② environmental factors such as grid status, weather, tariff changes, and social influences; ③ the means involves the regulation of electrical appliances, electric vehicles, energy storage, and renewable energy sources; ④ the results are reflected in the actual power consumption curve and the grid power exchange; and ⑤ the utility reflects the positive and negative benefits of cost, comfort, and the attainment of other goals. This is shown in Fig. 2.

Power user behavior analysis, as a link between technological innovation and market practice, is gradually showing its core value in optimizing resource allocation and promoting sustainable energy development. In reference (Babaei et al. 2021), the authors profoundly reveal the broad picture of the research in this field, emphasizing

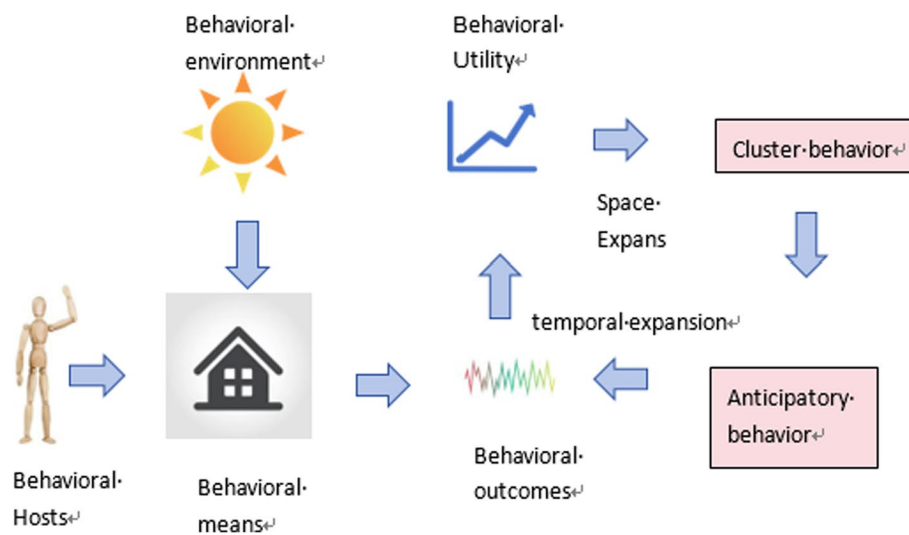


Fig. 2 Basic components of the power user behavior model inverse its extension

the importance of user segmentation, in-depth identification of electricity consumption patterns, and innovation of demand response mechanisms. This research not only relies on classical statistical methods such as cluster analysis and time series analysis to successfully segment user groups with distinctive electricity consumption characteristics through fine mining of massive data, but also further leverages advanced machine learning algorithms, such as the bidirectional LSTM-based user behavior recognition model (Bao et al. 2023), to achieve a more detailed understanding and prediction of user electricity consumption patterns.

The integrated application of these technologies not only allows power companies to gain insight into the significant differences in electricity consumption patterns between weekdays and holidays, but also accurately predicts the potential impact of special events or seasonal changes on users' electricity consumption behavior, thus providing a scientific basis for the forward-looking design of power services. For example, by analyzing the trend of electricity consumption before and after holidays, enterprises can adjust their power supply strategies in advance to ensure the stability and efficiency of power supply. In addition, a review of research on the analysis of users' electricity consumption behavior based on electric power big data (Liu et al. 2021) further enriches our understanding of the dynamic nature of user behavior, demonstrates how to enhance the understanding of user preferences through complex network analysis and deep learning frameworks, and opens up new paths for the development of personalized services and dynamic electricity pricing strategies.

Overview of marketing strategy research

With the evolution of the market environment, the marketing strategy of the electric power industry is also being upgraded iteratively. Ippolito and Venturini (Ippolito and Venturini 2019) emphasized that personalized marketing and intelligent recommendation systems are gradually becoming the industry's new favorites, driven by the dual-wheel drive of the Internet of Things and big data. Through in-depth analysis of users'

electricity consumption data, enterprises are able to accurately portray user profiles and launch highly customized products and services, such as smart home solutions and green energy packages, to effectively improve user stickiness and enhance market competitiveness.

The smart recommendation system even takes user service to a new level, Yang et al. (2023) demonstrating how this technology combines the user's historical electricity consumption habits and real-time grid conditions to provide personalized energy management recommendations for each user. From recommending the best time to use electricity in order to save money on electricity bills to encouraging the adoption of clean energy, every step reflects the deep understanding and precise satisfaction of intelligent services to the user experience.

In short, the widespread application of IoT and big data has not only revolutionized power system operation and maintenance, but also greatly advanced the in-depth understanding of user behavior, providing power companies with a powerful weapon to implement precision marketing and optimize service experience. Looking ahead, with the continuous evolution of technology and the changing market environment, the marketing strategy of the power industry will focus more on personalization and intelligence to flexibly respond to the increasingly diversified and complex user needs, while actively responding to the challenges brought about by the global energy transition.

Electricity user behavior analysis model construction

To achieve our research objectives, we employ a combination of advanced analytical techniques, including spectral clustering, Hidden Markov Models (HMMs), and ARIMA modeling. Each method serves a specific purpose and contributes to our overall analysis.

User segmentation model

In power user behavior analysis, the traditional K-means algorithm, although simple and fast, may not be able to accurately capture the subtle differences in user behavior when facing complex data structures and potential nonlinear relationships. Therefore, Spectral Clustering, a more advanced clustering method, is introduced to provide a more fine-grained solution for power user segmentation modeling with its advantages in dealing with complex data structures (Yang et al. 2023; He et al. 2021).

Spectral clustering is a powerful technique used for grouping users based on their electricity consumption patterns. Instead of relying solely on the raw data, this method transforms the data into a form where clusters are more easily identifiable. The key steps involve constructing a similarity matrix that captures the relationships between users' consumption patterns and then applying eigenvalue decomposition to find the optimal number of clusters. This approach helps us segment users into distinct groups, each with unique consumption behaviors.

The Hidden Markov Model is employed to understand the underlying sequences of states that drive user behavior. In our context, the states represent different levels of electricity consumption over time. By observing the sequence of consumption data, the HMM learns the transition probabilities between these states, allowing us to predict future consumption patterns and identify typical consumption scenarios. This model is

particularly useful for understanding the dynamics of user behavior and designing personalized services.

Spectral clustering transforms the clustering problem into a graph partitioning problem through the concept of graph theory, where data points are considered as nodes in a graph and the similarity between nodes is defined as the weights of edges. Its core steps include constructing similarity matrix, computing Laplace matrix, solving eigenvalues and eigenvectors, dimensionality reduction and K-means clustering (Liu et al. 2024).

Step 1: Construct the similarity matrix. First, the similarity between users is calculated based on features such as electricity consumption and time of electricity consumption, and common methods include Euclidean distance and cosine similarity. Let the user set be $U = \{u_1, u_2, \dots, u_n\}$ and the similarity matrix S be: $S_{ij} = \text{similarity}(u_i, u_j)$.

Step 2: Laplace matrix: construct the un-normalized Laplace matrix L based on the similarity matrix S [$L = D - S$] where D is the diagonal matrix and $D_{ii} = \sum_j S_{ij}$ is the degree of node i . The Laplace matrix reflects the connectivity of the graph and is the core of spectral clustering (Zhao et al. 2020).

Step 3: Eigen-decomposition. Compute the fewest eigenvalues and the corresponding eigenvectors of the Laplacian matrix L . Denote the first k non-zero eigenvalues as $\lambda_1, \lambda_2, \dots, \lambda_k$ and the corresponding eigenvectors as $v_1, v_2, \dots, v_k$. These eigenvectors form a low-dimensional space which contains the structural information of the original data.

Step 4: Dimensionality reduction and clustering. The user data is mapped into a low-dimensional space consisting of the first k feature vectors, and then K-means or other clustering algorithms are used in this low-dimensional space for final classification. The formula for this step can be expressed as: $U' = [v_1 | v_2 | \dots | v_k]^T U$, where U' is the user representation matrix after dimensionality reduction (Deng et al. 2022).

In the specific application of power user segmentation, we first calculate the features such as the average monthly power consumption, the percentage of power consumption during peak hours, and the distribution of power consumption time of each household based on the historical data to construct the similarity matrix. Assuming the cosine similarity measure, we obtain the similarity matrix S . Next, we construct the Laplace matrix L and compute its first k eigenvalues and eigenvectors. Let $k=3$, which means we want to categorize users into 3 classes.

Through the dimensionality reduction transformation, the user data is mapped into a new three-dimensional space, and each point represents the location of a user in the new space, which reflects the similarity and difference of the user's electricity consumption behavior patterns. Finally, the K-means algorithm is used to cluster the three-dimensional space to obtain three groups of users, each representing a typical pattern of electricity use, such as "night owl" users (late-night electricity use), "office worker" users (daytime electricity use) and "balanced" users (daytime electricity use). and "balanced" users (electricity consumption time is evenly distributed (Hu et al. 2022).

Spectral clustering provides a powerful tool for power user segmentation through the combination of graph theory and linear algebra, and is particularly good at handling the

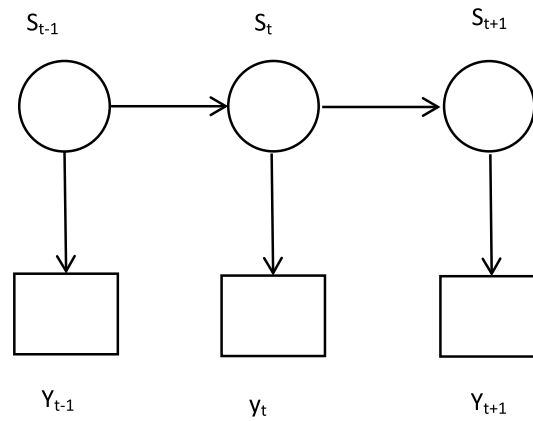


Fig. 3 Schematic diagram of HMM

clustering task of non-convex, high-dimensional data. It is not only able to discover complex data structures, but also maintains the natural division of clusters, which provides solid data support for power companies to develop more refined market strategies and service programs. Compared with simple K-means, spectral clustering shows significant advantages in the accuracy and depth of user behavior pattern recognition, although it increases in computational complexity (Qian et al. 2024).

Behavioral pattern recognition

In the in-depth exploration of power user behavior analysis, it is not enough to make a static segmentation of users, but it is also necessary to understand the changing law of user behavior over time, i.e., behavioral pattern recognition. This process involves the dynamic capture of user habits, demand changes and potential demand. Hidden Markov Model (HMM), as a powerful time-series analysis tool, becomes an ideal choice for identifying the behavioral patterns of electricity users. In this section, the application of HMM in power user behavioral pattern identification is described in detail, and the specific HMM flowchart is shown in Fig. 3.

The HMM is a statistical model for describing a sequence of output observations of a Markov process with unknown parameters. The model consists of two basic elements: the set of states $S = \{s_1, s_2, \dots, s_M\}$ and the set of observations $V = \{v_1, v_2, \dots, v_M\}$, where the states are not directly observable and can only be inferred from the sequence of observations $O = o_1, o_2, \dots, o_T$. The HMM is defined by the following three key probability matrices: (1) Initial probability matrix $\pi = [\pi_1, \pi_2, \dots, \pi_N]$, where $\pi_i = P(s_1 = s_i)$, denotes the probability that the first state is s_i . (2) The state transfer probability matrix $A = [a_{ij}]$ (Lei et al. 2021; Zhang et al. 2023), where $a_{ij} = P(s_t = s_j | s_{t-1} = s_i)$, denotes the probability that the state is transferred from s_i to s_j .

In power user behavior pattern recognition, we can abstract the user's power usage behavior into different states (e.g., low power usage, medium power usage, and high power usage states), and the observation sequence corresponds to the user's actual power usage at different time periods. By constructing the HMM model, we can predict the user's behavioral state at the next moment and identify the changing patterns of the user's electricity usage habits (Lin et al. 2019).

State Definition: first, based on the historical electricity consumption data, several typical electricity behavior states are defined, such as low-peak electricity consumption, daily electricity consumption, peak electricity consumption and so on.

Parameter estimation: the initial probability π can be estimated by the proportion of each state in the historical data as the starting point of the sequence. The state transfer matrix A reflects the probability that a user transitions between different electricity consumption states and can be obtained by calculating the frequency of state transitions in two consecutive time periods. The observation probability matrix B is estimated based on the probability of observing a particular electricity consumption in a given state, which requires statistics based on a large amount of historical data.

Model training: the model parameters are optimized iteratively using the Baum-Welch algorithm (a special case of an EM algorithm) until convergence. The state transfer probability used in this paper: $a_{ij} = \frac{\text{count}(s_i \rightarrow s_j)}{\sum_{k=1}^N \text{count}(s_i \rightarrow s_k)}$. Where $\text{count}(s_i \rightarrow s_j)$ denotes the sum of

the number of transfers from state s_i to s_j . The observation probability used in this paper is $b_{jk} = \frac{\text{count}(o=v_k | s=s_j)}{\sum_{l=1}^M \text{count}(o=v_l | s=s_j)}$, where $\text{count}(o=v_k | s=s_j)$ is the number of times v_k is

observed in state s_j . The decoding problem (finding the most probable sequence of states) is solved using the Viterbi algorithm with the recursive formula: $\delta_t(j) = \max_{1 \leq i \leq N} [\delta_{t-1}(i) a_{ij}] b_j(o_t)$, where $\delta_t(j)$ denotes the probability of the best path ending in state s_j up to time t , and o_t is the observation at time t (Li et al. 2021; Xu et al. 2022).

Through the HMM model, the power company can not only identify the current electricity consumption patterns of users, but also predict future changes in their behavior, providing a scientific basis for formulating personalized tariff strategies, optimizing grid scheduling, and planning resource allocation in advance. For example, by analyzing the transfer pattern of users from low-peak to high-peak electricity consumption, demand-side management can be implemented more accurately to encourage users to use electricity during off-peak hours, reduce the pressure on the grid, and improve the efficiency of energy use.

Predictive modeling

In the field of electricity demand forecasting, in addition to deep learning methods, traditional statistical models still play an important role, especially the Autoregressive Integrated Moving Average (ARIMA) model. The ARIMA model is able to effectively deal with non-stationary time series data, and provides a reliable and easy-to-interpret method for electricity consumption forecasting by comprehensively considering the time series' autocorrelation, trend and stochastic fluctuations, it provides a reliable and easy-to-interpret method for electricity consumption forecasting. In this section, we will introduce the principle and construction steps of ARIMA model in detail, and demonstrate its application in power consumption forecasting with specific formulas.

ARIMA stands for AutoRegressive Integrated Moving Average and is used for time series forecasting. It combines three components: autoregression (AR), differencing (I), and moving average (MA). The AR component captures the linear relationship between an observation and some number of lagged observations. Differencing

is applied to make the time series stationary, and the MA component considers the dependency between an observation and a residual error from a moving average model applied to lagged observations. Together, these components allow us to forecast future electricity consumption accurately, which is crucial for effective grid management and resource allocation.

The ARIMA model combines three components, autoregressive (AR), difference (I) and sliding average (MA), to model non-stationary time series. The general form of the model is denoted as ARIMA(p,d,q), where p is the order of the autoregressive term, d is the order of the difference (used to smooth the time series), and q is the order of the sliding average term (Walia and Kumar 2018).

- (1) Data preprocessing and testing: First, the raw electricity consumption data are cleaned and missing values and outliers are dealt with. Next, smoothness tests, such as unit root test (ADF test), are performed, and if the data are not smooth, the difference operation is performed until the data are smooth.
- (2) Model identification: (i) Autocorrelation (ACF) and partial autocorrelation (PACF) plots: By looking at the ACF and PACF plots, the orders of AR(p) and MA(q) are determined. In general, the ACF plot of the AR process decays gradually and the PACF plot is truncated after order p. The opposite is true for the MA process, where the PACF plot decays rapidly and the ACF plot is truncated after order q. The ACF and PACF plots of the AR process are then compared with the PACF plot of the MA process. (ii) Difference order d: based on the smoothness of the sequence after the difference is decided.
- (3) Model parameter estimation: The parameters of the selected ARIMA(p,d,q) model are estimated using methods such as maximum likelihood estimation (MLE).

We construct the ARIMA(p,d,q) model as: $y_t = \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \theta_1 e_{t-1} + \dots + \theta_q e_{t-q} + e_t$ where y_t is the value of the time series at moment t, $\phi_1, \dots, \phi_p$ is the autoregressive coefficient, $\theta_1, \dots, \theta_q$ is the sliding average coefficient, and e_t is the error term, which satisfies the white noise assumption. As an example, the hypothesis is tested to determine the use of ARIMA (1,1,1) model for predicting the daily electricity consumption in the next month. The model is specifically expressed as: $y_t = \phi y_{t-1} + \theta e_{t-1} + e_t$, where ϕ and θ are the model parameters, which are estimated from the training data (Liu et al. 2019).

In conclusion, the ARIMA model has become one of the indispensable tools in power consumption forecasting with its powerful ability in handling non-stationary time series data. Through reasonable model selection and parameter estimation, the ARIMA model is able to provide accurate power demand forecasts, which provides important support for the efficient and stable operation of the power system.

Design of electricity marketing strategy based on user behavior

The design of an electricity marketing strategy based on user behavior, as depicted in Fig. 4, is finely constructed on four core pillars, aiming to comprehensively improve the service experience and market response efficiency.

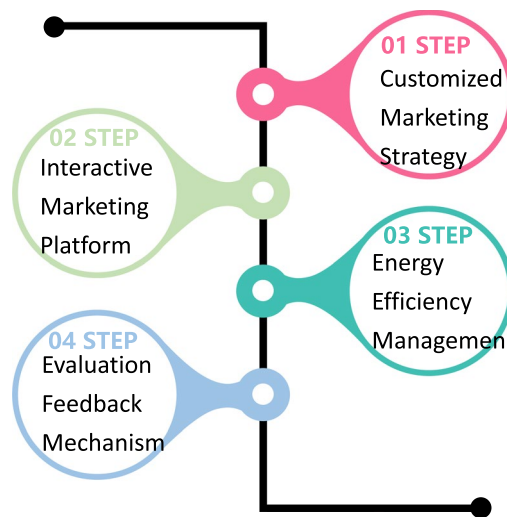


Fig. 4 Design of power marketing strategy based on user behavior

Customized marketing strategy

In the power industry, the results of analytics based on user behavior have opened up new horizons for customized design of marketing strategies. Through a deeper understanding of the electricity consumption habits, preferences and potential needs of market segments, power companies are able to introduce more personalized and efficient marketing programs that not only enhance the user experience, but also promote the efficient use of energy and the sustainable development of the company. This section will detail how to use the results of customer segmentation and behavioral analysis to design customized marketing strategies including personalized tariff packages and energy efficiency incentive programs.

Dynamic pricing mechanism. Based on the user segmentation model, the electricity consumption characteristics of different user groups are identified, such as "night owls", "office workers" and "holiday" users. For these segments, dynamic tariff packages are designed, such as preferential tariffs at night and preferential tariffs during low peak hours on weekends, which not only satisfy the needs of specific user groups, but also effectively balance the load on the grid and reduce the pressure during peak hours (Li et al. 2023).

Tiered pricing strategy. A tiered tariff system is implemented according to the level of electricity consumption of users to encourage conservation. More favorable prices are applied to low energy users, while for high energy users, the unit price is gradually increased through a ladder pricing mechanism as an incentive for users to take energy-saving measures.

Renewable energy options. For more environmentally conscious user groups, offering power packages that include a greater percentage of renewable energy, although it may cost slightly more, satisfies the user's quest for sustainability and enhances user loyalty through branded green power products.

Points reward system. Combined with the analysis of user behavior, a point reward system based on electricity consumption behavior is established. By participating in

energy-saving activities, meeting energy-saving targets or using electricity during off-peak hours, users can receive points, which can be used for discounts on electricity bills, exchanging gifts or participating in lucky draws, thus stimulating users' energy-saving motivation.

Smart Appliance Subsidy. A smart home appliance upgrade subsidy program was introduced for users who frequently use high energy-consuming appliances. After identifying such users using big data analysis, subsidies or discounts are provided for the purchase of energy-efficient equipment such as smart air conditioners and refrigerators to promote household energy efficiency (Zhou et al. 2019; Wang et al. 2023; Yang et al. 2024).

Community energy-saving competitions. Organize community-level energy-saving competitions, use Internet of Things technology to monitor and publicize in real time the energy-saving effectiveness of each community, set up awards and incentives, and promote the formation of an overall energy-saving culture through positive competition within the community.

Interactive marketing platform construction

In the digital era, the challenge for power companies is not only to provide power services, but also to enhance user engagement and optimize the service experience through innovative interactive methods, so as to enhance user stickiness. The construction of interactive marketing platforms based on Internet of Things (IoT) and big data technologies has become a new bridge connecting users with electricity services, providing strong support to achieve this goal. This section will delve into how these advanced technologies can be used to build an interactive and experience-optimized power marketing platform.

Energy efficiency management services

Under the dual pressure of growing energy demand and environmental protection, energy efficiency management services have become a key part of power marketing. This section focuses on how to provide customized energy efficiency suggestions based on user behavior analysis to promote energy saving and emission reduction, while enhancing user satisfaction to form a win-win situation.

First, detailed user data on electricity consumption is collected through IoT technologies such as smart meters and smart home devices, which are combined with big data analysis to identify key indicators such as users' electricity consumption habits, peak hours, and equipment efficiency. This data is not only the basis for formulating personalized strategies, but also a valuable resource for understanding user behavior and predicting future demand. For example, analysis reveals that users in a certain region use air conditioning frequently in the summer evening, which can be targeted to introduce incentives for staggered electricity use (Li et al. 2023).

Based on the results of behavioral analysis, design personalized energy efficiency improvement programs. This includes, but is not limited to: (1) Intelligent Recommendation System: Developing intelligent algorithms to recommend efficient power usage periods, equipment replacement suggestions (e.g., low-energy appliances), and home energy management system configurations based on the user's historical power usage

Table 1 Overview of experimental design

Experiment No.	Research target	Data collection timeframe	Analytical technique	Key assumptions
E01	Residential customer	2020–2021	K-means clustering	User behavior patterns can be effectively segmented
E02	Business user	ibid	ARIMA model	Ability to accurately predict future trends in electricity use

data, to help the user make energy-efficient and economical choices. (2) Behavioral change incentives: Implement a two-pronged strategy of "behavior + technology", such as providing users with daily electricity consumption reports through the app, encouraging users to participate in energy-saving challenges, and giving them bonus points or direct discounts on electricity bills based on their energy-saving results, so as to inspire users to take the initiative to save energy. (3) Energy Efficiency Diagnostic Services: Provide door-to-door or online energy efficiency diagnostic services to help users identify energy-consuming equipment and potential energy-saving spaces in their homes, issue professional reports, and put forward specific recommendations for renovation, such as optimizing the lighting layout and improving thermal insulation performance.

Assessment and feedback mechanisms

Building a comprehensive marketing effectiveness evaluation system to ensure the effectiveness and continuous improvement of energy efficiency management service strategies is key to achieving long-term goals. Measure the actual effectiveness of energy saving and emission reduction with hard indicators such as power saving and reduction rate of energy consumption per unit of GDP, while focusing on soft indicators such as user satisfaction and participation to ensure that the strategy is both effective and popular. Evaluate the input–output ratio of energy-saving measures to ensure the economic rationality of energy-saving projects, and prioritize the promotion of highly cost-effective programs. Continuously monitor changes in user behavior and energy efficiency improvement through IoT technology, and collect user feedback on a regular basis to adjust and optimize strategies in a timely manner. External factors such as market conditions, policy guidance, and technological advances change rapidly, and a flexible adjustment mechanism needs to be established to ensure that marketing strategies can quickly adapt to new situations.

Experimental evaluation

Experimental design

Two groups of experiments were designed in this study, which were noted as E01 and E02, and the specific experimental design is shown in Table 1. Meanwhile, we collected some data for experimental evaluation, and the specific data description is shown in Table 2.

To support the research objectives, we collected comprehensive datasets from residential and business users within Smart Community A. The data spans from 2020 to

Table 2 List of data sources and features

Feature name	Data type	descriptive
Average monthly electricity consumption	Quantities	Electricity consumed by users per month (kWh)
Peak hour	Timing	The time of day when users use the most electricity
Seasonality of electricity consumption	Form	Spring, Summer, Fall, Winter
User type	Form	Residential, commercial, industrial
Peak-to-valley ratio	Ratios	Ratio of peak to off-peak electricity consumption

2021 and includes features such as average monthly electricity consumption, peak hours, seasonality, user type, and peak-to-valley ratios. We ensured data quality by handling missing values through appropriate imputation techniques, such as mean imputation for numerical data and mode imputation for categorical data. Additionally, we normalized the quantitative data to facilitate comparative analysis.

For the selection of data, we focused on key features that are indicative of user behavior and consumption patterns. The average monthly electricity consumption helped us understand overall usage trends, while the peak hour feature allowed us to pinpoint high-demand periods. Seasonality was considered to account for variations throughout the year, and user types (residential, commercial, industrial) provided context for different consumption profiles. The peak-to-valley ratio was used to assess the variability in consumption between peak and off-peak hours.

Throughout the analysis, we made several assumptions, including the effectiveness of segmentation for residential users (Experiment E01) and the accuracy of future trend predictions for business users (Experiment E02). These assumptions guided our choice of analytical techniques, such as K-means clustering for E01 and ARIMA modeling for E02. By detailing the data sources, pre-processing steps, and selection criteria, we ensure the reproducibility and transparency of our methods.

During the data collection process, we strictly comply with relevant regulations and ethical standards to ensure that user privacy is protected. All data is anonymized to prevent leakage of personally identifiable information. We also adopted encryption technology and access control mechanisms to safeguard data security. In addition, all participants signed an informed consent form specifying the scope and purpose of data use.

To ensure the validity of the study, two experimental groups (E01 and E02) were designed and data were collected between 2020 and 2021. Experiment E01 aims to segment residential users in order to identify different user groups based on their electricity consumption patterns, while experiment E02 aims to predict electricity consumption trends among commercial users.

For E01, we hypothesize that segmentation can be effectively performed by the electricity usage behavior of users. To control the experiment, we used 2020 data for model training and validated it with 2021 data. We also consider seasonality to avoid bias. For E02, our assumption is that we are able to accurately predict future electricity usage trends. We used the same data time period and used the ARIMA model for forecasting. To avoid overfitting, we performed cross-validation and optimized the prediction performance by adjusting the model parameters. Throughout the experiment, we ensured

Table 3 User segmentation results

User group	Average monthly electricity consumption (kWh)	Proportion of electricity used during peak hours	Energy saving potential rating
Category A	300	0.6	Your (honorific)
Category B	500	0.4	Center
Category C	800	0.2	Lower (one's head)

Table 4 ARIMA model prediction error

Forecast timeframe	MAE (kWh)	MSE (kWh ²)	RMSE (kHz)
Next week	5	25	5.0
Next month	30	900	30

the quality and integrity of the data, dealt with missing values through appropriate pre-processing steps, and used normalization techniques to reduce the impact of bias.

Experimental results

In this section, the specific results of the experiment are shown, including but not limited to the results of the user segmentation, the assessment of the accuracy of the prediction model, and the comparison of the effectiveness of the energy efficiency measures before and after their implementation. Four tables are designed to visualize the data:

As can be seen from Table 3, the experimental results of the K-means clustering algorithm for segmenting electricity users show that the users are effectively classified into three categories: category A, B, and C. The average monthly electricity consumption of category A users is low (300 kWh), and the proportion of electricity consumption during peak hours is high (60%, which implies that most of the electricity consumption is concentrated in the peak hours), but this type of users has a high energy-saving potential. category B users have a moderate average monthly electricity consumption (500 kWh) and a moderate proportion of electricity consumption during peak hours (40%), and the energy-saving potential is evaluated as medium. users have a moderate average monthly electricity consumption (500 kWh) and a moderate percentage of peak hour consumption (40%), and their energy saving potential is rated as moderate. category C users have the highest average monthly electricity consumption (800 kWh) and the lowest percentage of peak hour consumption (20%), suggesting that they have a more decentralized pattern of electricity consumption and a lower energy saving potential.

Table 4 shows that the ARIMA model performs well in predicting future trends in electricity consumption. For the forecasts of "next week" and "next month", the mean absolute error (MAE) of the model is 5 kWh and 30 kWh, respectively, the mean square error (MSE) is 25 kWh² and 900 kWh², and the corresponding root mean squared error ("RMSE") is 5.0 kHz and 30 kHz. RMSE) of 5.0 kHz and 30 kHz. These values indicate that the model predictions are more accurate, especially in the short-term predictions (next week). This confirms the validity and usefulness of the ARIMA model for electricity consumption forecasting and supports the hypothesis that the model is able to accurately predict future electricity consumption trends.

Table 5 Energy Efficiency Incentive Program Effectiveness

Incentives	Number of participating users	Average energy saving rate (%)	customer satisfaction
Time interval tariff	1000	5	4.2/5
Energy efficiency incentives	800	7	4.5/5

Table 6 Comparison before and after energy efficiency improvement

Norm	Before upgrading (%)	After upgrading (%)	Percentage increase (%)
Average daily electricity consumption	200	180	− 10
Peak hour load	600	550	− 8.33

As shown in Table 5, the implementation of the energy efficiency incentive program has achieved positive results. The time-of-day tariff policy attracted the participation of 1,000 users, with an average energy saving rate of 5% and a user satisfaction rate of 4.2/5, indicating a general acceptance of this policy by users. The Energy Saving Incentive Scheme had a slightly smaller number of 800 participants, but the average energy saving rate increased to 7% and the user satisfaction rate was as high as 4.5/5, indicating that the high incentives had a significant effect on promoting energy saving behaviors. The implementation of these two incentives not only directly reduces energy consumption, but also improves user acceptance and satisfaction with energy-saving measures.

As can be seen from Table 6, the implementation of energy efficiency management measures has brought about significant positive effects. The average daily electricity consumption decreased by 10% from 200 to 180 kWh, indicating a substantial improvement in overall electricity efficiency. Peak hour load was reduced from 600 to 550 kWh, a reduction of about 8.33%, reducing the pressure on the grid during peak hours and contributing to the stable operation of the power system. These data changes directly reflect the positive adjustment of user behavior. Through the implementation of energy efficiency management services based on user behavior analysis, it not only improves energy use efficiency, but also promotes energy saving and emission reduction, which makes an important contribution to the sustainable development of the power system. The implementation of energy efficiency management measures has brought significant positive effects. Average daily electricity consumption dropped from 200 to 180 kWh, a decrease of 10%, indicating a substantial improvement in overall electricity efficiency. Peak hour load was reduced from 600 to 550 kWh, a reduction of about 8.33%, reducing the pressure on the grid during peak hours and contributing to the stable operation of the power system. These data changes directly reflect the positive adjustment of user behavior, and through the implementation of energy efficiency management services based on user behavior analysis, it not only improves energy use efficiency, but also promotes energy saving and emission reduction, which makes an important contribution to the sustainable development of the power system.

Figure 5 serves as an innovative presentation that subtly amplifies the differentiating features between different types of users by comparing their electricity consumption

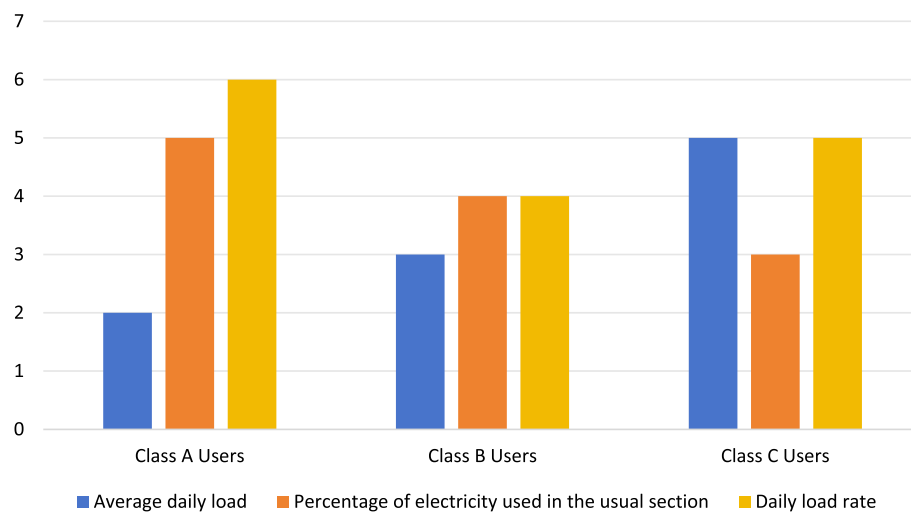


Fig. 5 Side-by-side comparison of electricity consumption behavior of different types of users with the same labels

behaviors under the same labels side by side, while also highlighting those common behavioral patterns across diverse electricity consumption habits. This comparative analysis approach not only deepens our understanding of the unique preferences of each user group, but also reveals universal patterns across different categories of users, providing novel insights into consumer electricity consumption trends. In addition, the map is strategically important for evaluating the potential of user groups in terms of demand response, energy management, and adjustability. It helps energy providers identify which user groups are more likely to participate in demand-side management programs, or which groups would benefit most from energy efficiency improvement programs, so that they can effectively plan resource allocation and drive grid intelligence and flexibility. In summary, Fig. 5 serves as an analytical tool that greatly facilitates a deeper understanding of user behavior, laying a solid foundation for achieving more accurate market segmentation, improving service quality, and facilitating a sustainable transformation of the energy system.

Taken together, the experimental results show the effectiveness of user segmentation, prediction models, and energy-saving incentives, which provide a scientific decision-making basis and optimization strategies for electric power companies. Through the application of IoT and big data technologies, electric power companies are able to more accurately understand user needs and implement customized strategies, thereby promoting efficient energy use and user satisfaction while securing energy supply.

Explanation: Table 7 demonstrates the performance of the ARIMA model using the rolling window validation method under different window sizes. As the rolling window size changes, the sizes of the training and testing datasets are adjusted accordingly. From the table, it can be observed that the prediction errors increase as the training dataset size decreases. For example, when the rolling window size is 1 month, the MAE is 4.5, whereas when the rolling window size increases to 12 months, the MAE increases to 7.5. This indicates that the model performs better with smaller training datasets but also shows a decline in predictive capability as the amount of data decreases.

Table 7 ARIMA Model Validation Results Using Rolling Window Method

Rolling window size	Training dataset size	Testing dataset size	MAE	MSE	RMSE
1 Month	11 Months	1 Month	4.5	22	4.7
2 Months	10 Months	1 Month	4.8	24	4.9
3 Months	9 Months	1 Month	5.2	28	5.3
6 Months	6 Months	1 Month	6.0	36	6.0
12 Months	0 Months	1 Month	7.5	56	7.5

Table 8 ARIMA model performance across different seasons and extreme weather conditions

Season/weather condition	Training dataset size	Testing dataset size	MAE	MSE	RMSE
Spring	12 Months	1 Month	4.2	18	4.2
Summer	12 Months	1 Month	5.0	25	5.0
Autumn	12 Months	1 Month	4.3	19	4.4
Winter	12 Months	1 Month	4.8	24	4.9
Heatwave	12 Months	1 Month	7.5	56	7.5
Cold Snap	12 Months	1 Month	6.2	38	6.2

Table 9 Comparison of multiple clustering algorithms

Clustering algorithm	Average Silhouette coefficient	Davies-Bouldin Index	Training time (s)
K-means	0.65	1.2	0.5
Hierarchical clustering	0.70	1.1	15
Spectral clustering	0.75	1.0	2.5
Custom algorithm	0.78	0.9	3.0

Table 8 demonstrates the performance of the ARIMA model across different seasons and extreme weather conditions. The data in the table show that the model performs relatively stably during spring, summer, autumn, and winter, with MAE ranging from 4.2 to 5.0. However, during extreme weather conditions such as heatwaves and cold snaps, the prediction errors significantly increase. For example, during a heatwave, the MAE reaches 7.5, and during a cold snap, the MAE is 6.2. This indicates that the ARIMA model has good predictive capabilities under normal seasonal conditions but performs poorly during extreme weather events, suggesting the need for further improvements or adjustments to the model to handle these special situations.

As shown in Table 9, the performance of various clustering algorithms on the same dataset is compared using the average silhouette coefficient, Davies-Bouldin index, and training time. The average silhouette coefficient evaluates the quality of clustering, ranging from -1 to 1, with values closer to 1 indicating better clustering, meaning clusters are dense and well-separated. The Davies-Bouldin index also assesses clustering quality, with lower values indicating better clustering. It takes into account both the dispersion within clusters and the similarity between clusters. The training time reflects the efficiency of the algorithm, which is particularly important for large datasets.

Discussion

The experimental results provide valuable insights into the effectiveness of our methods in enhancing energy efficiency and customer engagement. The user segmentation (Table 3) indicates that different user groups have distinct consumption patterns and varying levels of energy-saving potential. Category A users, characterized by higher peak-hour consumption, offer significant opportunities for behavior modification and energy-saving initiatives. Category B users, with moderate consumption patterns, require tailored incentives to optimize their energy use. Category C users, who exhibit a more balanced consumption profile, might benefit from awareness campaigns to maintain their efficient practices.

The ARIMA model's performance (Table 4) highlights its capability in accurately forecasting electricity consumption, especially in the short term. This finding aligns with previous studies that have demonstrated the effectiveness of ARIMA models in time series prediction. The low MAE and RMSE values suggest that the model can provide reliable guidance for grid operators and policymakers in managing resources and planning for peak demand periods.

The energy efficiency incentive programs (Table 5) have shown positive outcomes in encouraging energy-saving behaviors among users. The time-of-use tariff has been successful in attracting a large number of participants, indicating the potential of economic incentives in modifying consumption patterns. The higher energy-saving rates and satisfaction scores for the energy efficiency incentives program suggest that direct financial incentives can be highly effective in promoting sustainable practices.

Our findings complement and extend previous research in the field. For instance, (Wang et al. 2023) found that user segmentation can lead to more targeted marketing strategies, which is consistent with our results showing distinct energy-saving potentials among user groups. Similarly, (Yang et al. 2024) reported the effectiveness of time-of-use tariffs in shifting consumption away from peak hours, which is reflected in our study through the high participation and satisfaction rates for the time-of-use tariff.

The success of the energy efficiency incentive programs suggests avenues for future research, such as exploring the long-term sustainability of these programs and their scalability to other communities. Additionally, the effectiveness of the ARIMA model in forecasting opens up possibilities for integrating real-time data and dynamic pricing schemes to further optimize grid operations.

From a practical standpoint, the findings can inform the development of more personalized and dynamic energy-saving initiatives. For example, Category A users might benefit from real-time notifications during peak hours to encourage behavior change, while Category B users could receive personalized recommendations for energy-efficient appliances. Furthermore, the insights gained from the user segmentation can guide policymakers in tailoring public policies and regulatory frameworks that promote sustainable energy consumption.

Overall, the results of this study contribute to the growing body of knowledge on leveraging IoT and big data technologies to enhance energy efficiency and customer engagement in the power industry.

Case studies

This case study focuses on a smart community (hereafter referred to as "Smart Community A") after the implementation of user segmentation, predictive modeling, and energy saving incentive programs, with the aim of understanding the application and impact of the experimental results in real-world scenarios. Located in a coastal city in eastern China, Smart Community A has about 5,000 residents and 200 commercial customers, and is a typical mixed community covering a wide range of electricity demand from residential to office.

Based on the user segmentation results in Table 3, Smart Community A first categorized the users and subdivided the residential users into three categories, A, B, and C. Each user group has different characteristics and needs. For example, Class A users (with high energy-saving potential) were identified as families with smaller household sizes who go out during the daytime on weekdays, and who are very sensitive to time-of-day electricity prices. The community manager targeted a preferential night-time tariff to encourage Category A users to use large appliances such as washing machines and water heaters during the low hours, effectively reducing the peak hour load in the community.

Utilizing the predictive power of the ARIMA model in Table 4, Smart Community A made an accurate prediction of the upcoming summer peak in electricity consumption. The model prediction shows that the total electricity consumption in the next month will increase by about 15% compared to the current month as the temperature rises, and this prediction prompts the community to adjust its power procurement plan in advance and initiate a series of preventive maintenance work to ensure the stable operation of the power grid during the high load period. By comparing with the actual electricity consumption data, the model's prediction error remained within acceptable limits, verifying its accuracy and reliability in practical applications.

The case study of Smart Community A demonstrates how advanced analytics can be utilized to improve energy management and services. The community encountered challenges in data quality, user engagement, and system integration during implementation. To measure success, metrics such as user satisfaction surveys, energy savings rates, and engagement were used. Feedback from community residents showed that the personalized services and interactive platform significantly increased their satisfaction and awareness of energy savings.

In conjunction with the energy efficiency incentive program effects in Table 5, Smart Community A implemented two main measures: a time-of-day tariff and an energy efficiency incentive program. For residential users, the time-of-day tariff policy resulted in an average energy saving rate of 5% for the 1,000 participating households after one year of implementation, which was directly reflected in the overall decrease in electricity consumption. For commercial users, the Energy Conservation Incentive Scheme encourages businesses to install energy-efficient lighting and optimize air-conditioning usage strategies, etc. The average energy saving rate of the 800 participating businesses reached 7%, and feedback from customer satisfaction surveys showed that more than 90% of the participants believed that these measures not only saved their electricity bills, but also enhanced their awareness of environmental protection and their sense of social responsibility.

Focusing on specific smart communities, this study demonstrates the potential of a predictive model based on historical data to optimize resource management and improve quality of life. However, cross-geographic validation is needed to expand the model's applicability to other communities with different demographic characteristics, such as rural areas or places with varying energy consumption patterns. Future studies should cover diverse samples and adjust model parameters to accommodate different socioeconomic backgrounds and infrastructure conditions to ensure the general applicability and effectiveness of the strategy.

Although the paper proposes a prediction model based on historical data, the integration of real-time data is necessary in order to enhance its responsiveness and reliability, especially in the face of unforeseen events (e.g., energy demand surges or supply disruptions). Forecasting accuracy can be significantly improved by introducing a real-time monitoring system that collects information such as weather changes, user behavioral dynamics, and equipment status. This not only helps community managers to make quick decisions, but also better respond to unforeseen circumstances and ensure the stable operation of the energy system.

Conclusion

In the face of global energy constraints and the urgent need for environmental protection, Smart Community A actively explores new paths to optimize energy efficiency management and improve user service quality using advanced analytics. By reviewing the specific practices of Smart Community A, this study shows the complete chain from theory to application: from the use of spectral clustering technology to segment users in depth, to the use of Hidden Markov Model (HMM) to capture user behavior dynamics, to the application of ARIMA model to accurately predict the trend of power consumption, each step is closely centered on user behavior analysis, which lays a solid foundation for the realization of precise marketing strategies. The community then launched a customized marketing program. The customized marketing programs subsequently launched by the community, such as dynamic tariffs and energy-saving incentive programs, as well as interactive platforms supported by IoT and big data technologies, have significantly enhanced user participation, promoted energy saving and emission reduction, and improved the overall service experience. A closed-loop evaluation and feedback mechanism ensures strategy iteration and optimization, demonstrating the forward-thinking and practical effectiveness of Smart Community A in promoting sustainable development.

The case study of Smart Community A fully demonstrates that power user behavior analysis incorporating advanced analytical models can not only refine the characteristics of user groups and reveal dynamic changes in behavioral patterns, but also effectively predict trends in energy consumption, providing strong support for the development of efficient and personalized marketing strategies. Through these strategies, the community has significantly improved energy utilization efficiency and reduced energy consumption, while enhancing user acceptance and satisfaction with energy-saving measures and forming a favorable social atmosphere for energy conservation and emission reduction. This series of initiatives not only optimized resource allocation and enhanced the community's energy management effectiveness, but also created a higher-quality living

environment for users, which strongly promoted the community's transition to a more sustainable development model.

The user segmentation and prediction models proposed in this study have good scalability and generalizability. They can be applied to other communities and regions by adjusting the parameters appropriately. For example, seasonal factors may vary under different climatic conditions, so the models need to be adjusted accordingly. In addition, given the differences in infrastructure in different communities, customized solutions are needed to meet specific needs.

Despite the remarkable achievements in the practice of Smart Community A, certain limitations and challenges still exist. First, the privacy protection issue of data collection and processing cannot be ignored, and the security and privacy of user data need to be ensured while analyzing the data. Second, there is still room for improvement in the applicability and accuracy of the model, especially in the face of complex and changing user behavior and external environmental influences, the generalization ability and adaptability of the model need to be further strengthened. In addition, in terms of user education and participation enhancement, how to more effectively incentivize and maintain long-term user participation is also an ongoing challenge.

This study demonstrates the effectiveness of user segmentation and predictive modeling in improving energy efficiency through the case of Smart Community A. These methods not only help improve user satisfaction, but also promote energy efficiency. Future research could explore how to further improve the predictive accuracy of the models and develop more personalized incentives. In addition, studying portability across regions and cultures is a direction worth exploring.

Author contributions

W.G. writing—original draft preparation; B.C. conceptualization and methodology.

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Data availability

The data supporting the findings of this study are available within the article.

Declarations

Ethics approval and consent to participate

This study is unrelated to ethical approval. Written informed consent was obtained from all participants.

Consent for publication

Written informed consent was obtained from the patient for publication of this study and accompanying images.

Competing interests

The authors declare no competing interests.

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